```
$IfNot ParametricTable
i=1300
Q_Evap = 5
$EndIf
JuneAir_4_pow_gshp_gshp_ac = lookup('FromExcel', i, 'Work_all')
JuneAir_20_t_gshp_evap_out  = lookup('FromExcel', i, 'Evap_Out')
JuneAir_21_t_gshp_evap_in = lookup('FromExcel', i, 'Evap_in')
JuneAir_15_t_gshp_compsr_out = lookup('FromExcel', i, 'Comp_out')
JuneAir_14_t_gshp_hex_out_r32 = lookup('FromExcel', i, 'Cond_out')


R$ = 'R32'
Subcool = T[4]-T[5]
SHV = T[2]-T[1]

COP = (h[2]-h[6])/(h[3]-h[2])
r_v = P[3]/P[1]
Eta_comp = 0.8
Eta_comp = (h3s-h[2])/(h[3]-h[2])
"State 1 : Outlet Evap Sat.Vapor"
T[1] = T[6]
P[1] = P[6]
x[1] = 1
h[1] = enthalpy(R$,x=x[1],P=P[1])
"State 2 : Outlet Evap + SHV"
P[2] = P[1]
T[2] = JuneAir_20_t_gshp_evap_out
h[2] = enthalpy(R$,T=T[2],P=P[2])
s[2] = entropy(R$,T=T[2],P=P[2])
"State 3 : Outlet Compressor"
s[3]=s[2]
T[3] = temperature(R$,P=P[3],h=h[3])
P[3] = P[4]
h3s = enthalpy(R$,P=P[3],s=s[3])
"State 4 : Outlet Condenser"
T[4] = t_sat(R$,P=P[4])
P[4] = 3301.325
x[4] = 0
h[4] = enthalpy(R$,T=T[4],x=x[4])
"State 5 : Outlet Condenser+Subcool"
T[5] = JuneAir_14_t_gshp_hex_out_r32
P[5] = P[4]
h[5] = enthalpy(R$,T=T[5],P=P[5])
"State 6 : Inlet Evap"
T[6] = JuneAir_21_t_gshp_evap_in
P[6] = 941.325
h[6]=h[5]

"First law"
Q_Evap + m_dot*h[6] = m_dot*h[2]
m_dot*h[2] =m_dot*h[3] +W_Comp
Q_Cond + m_dot*h[3] = m_dot *h[5]
```

SOLUTION

Unit Settings: SI C kPa kJ mass deg

(Table 2, Run 100)

$COP = 4.076$

$\eta_{comp} = 0.8$

$h3s = 579.5$

$i = 1300$

$JuneAir_{14,t,gshp,hex,out,r32} = 28.61$

$JuneAir_{15,t,gshp,compsr,out} = 37.42$

$JuneAir_{20,t,gshp,evap,out} = 12.5$

$JuneAir_{21,t,gshp,evap,in} = 7.8$

$JuneAir_{4,pow,gshp,gshp,ac} = 0.88$

$\dot{m} = 0.01826$

$Q_{Cond} = -6.227$ [kW]

$Q_{Evap} = 5$ [kW]

R$ = 'R32'

$r_v = 3.507$

$SHV = 4.7$

$Subcool = 23.56$

$W_{Comp} = -1.227$ [kW]

No unit problems were detected.

Arrays Table: Main

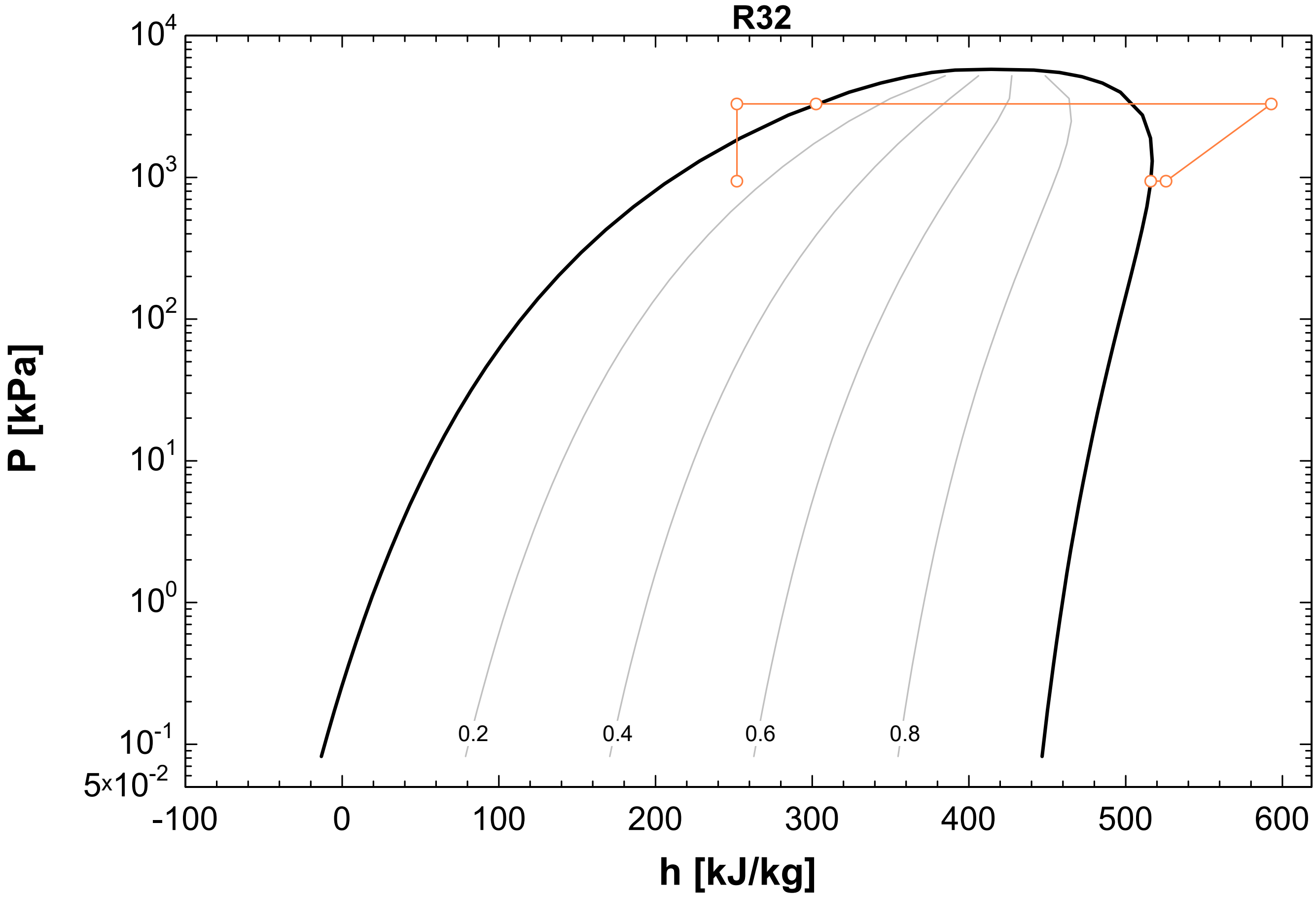
	T_i	P_i	x_i	h_i	s_i
1	7.8	941.3	1	516	
2	12.5	941.3		525.8	2.172
3	106.1	3301		593	2.172
4	52.17	3301	0	302.5	
5	28.61	3301		251.9	
6	7.8	941.3		251.9	

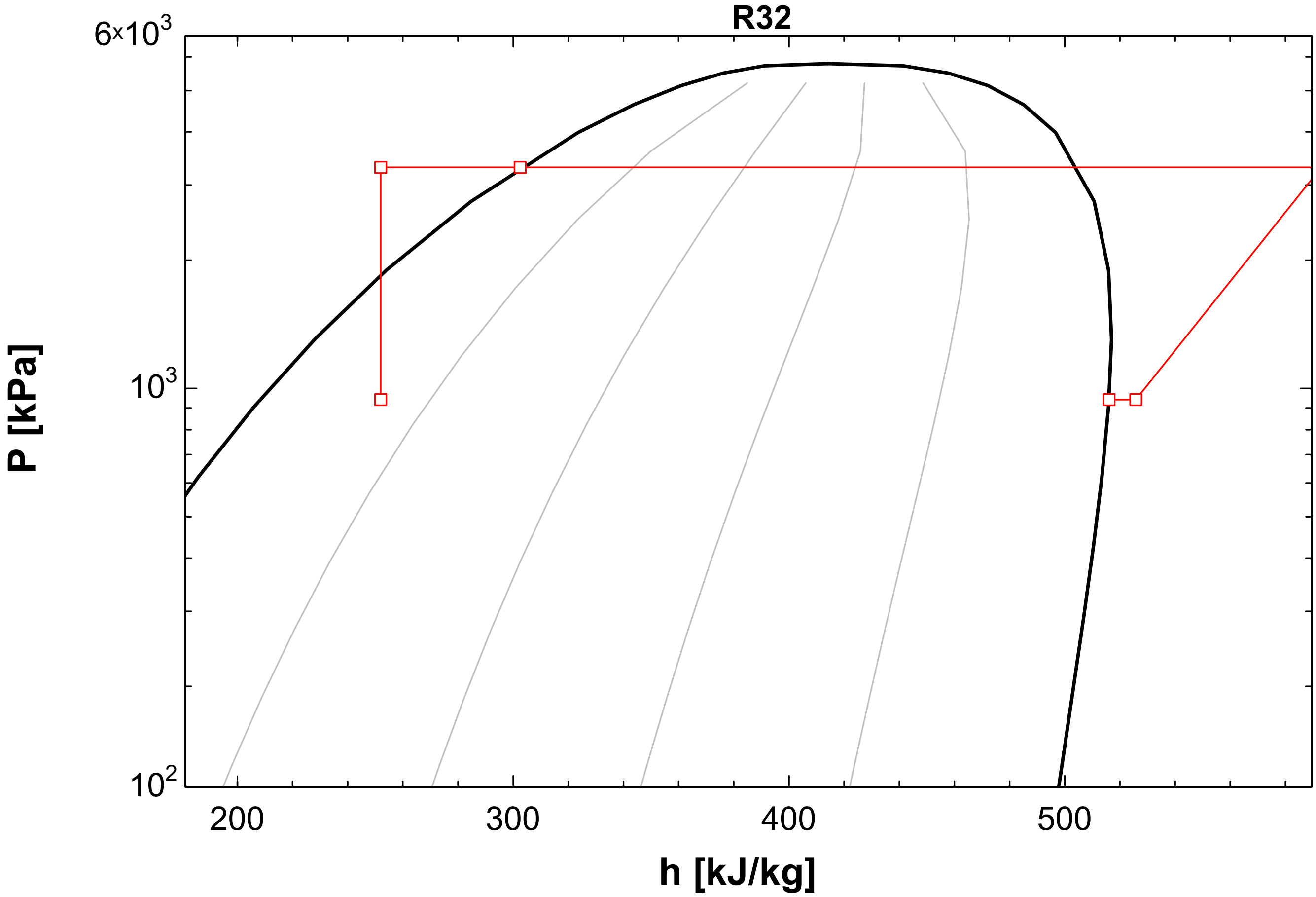
Parametric Table: Table 2

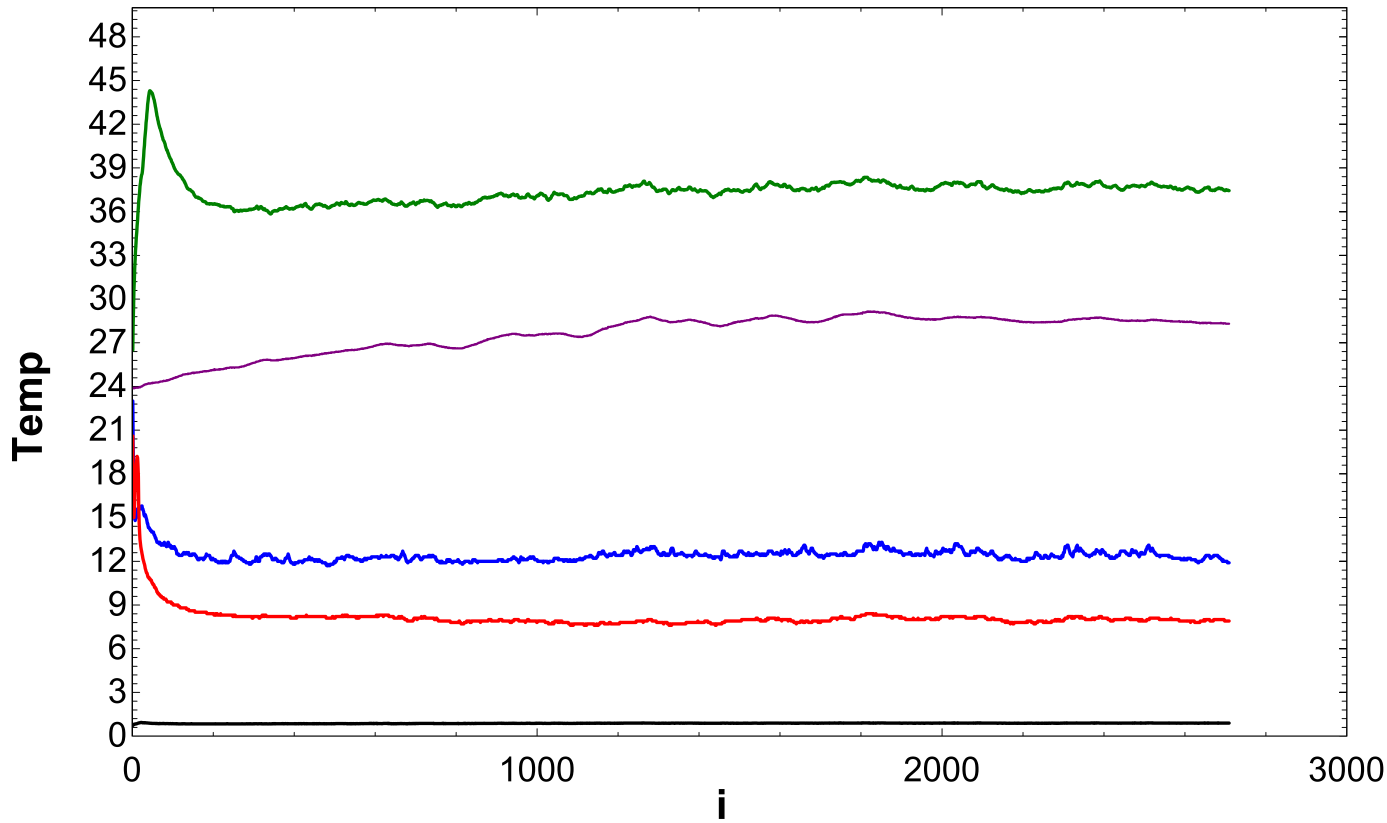
	i	$\dot{m}$	Q_{Cond} [kW]	Q_{Evap} [kW]	W_{Comp} [kW]
Run 1	1300	0.001826	-0.6227	0.5	-0.1227
Run 2	1300	0.001992	-0.6793	0.5455	-0.1338
Run 3	1300	0.002158	-0.7359	0.5909	-0.145
Run 4	1300	0.002324	-0.7925	0.6364	-0.1561
Run 5	1300	0.00249	-0.8491	0.6818	-0.1673
Run 6	1300	0.002656	-0.9057	0.7273	-0.1784
Run 7	1300	0.002822	-0.9623	0.7727	-0.1896
Run 8	1300	0.002988	-1.019	0.8182	-0.2007
Run 9	1300	0.003154	-1.076	0.8636	-0.2119
Run 10	1300	0.00332	-1.132	0.9091	-0.223
Run 11	1300	0.003485	-1.189	0.9545	-0.2342
Run 12	1300	0.003651	-1.245	1	-0.2453
Run 13	1300	0.003817	-1.302	1.045	-0.2565
Run 14	1300	0.003983	-1.359	1.091	-0.2676
Run 15	1300	0.004149	-1.415	1.136	-0.2788
Run 16	1300	0.004315	-1.472	1.182	-0.2899
Run 17	1300	0.004481	-1.528	1.227	-0.3011
Run 18	1300	0.004647	-1.585	1.273	-0.3122
Run 19	1300	0.004813	-1.642	1.318	-0.3234
Run 20	1300	0.004979	-1.698	1.364	-0.3345
Run 21	1300	0.005145	-1.755	1.409	-0.3457
Run 22	1300	0.005311	-1.811	1.455	-0.3568
Run 23	1300	0.005477	-1.868	1.5	-0.368
Run 24	1300	0.005643	-1.925	1.545	-0.3791
Run 25	1300	0.005809	-1.981	1.591	-0.3903
Run 26	1300	0.005975	-2.038	1.636	-0.4014
Run 27	1300	0.006141	-2.094	1.682	-0.4126
Run 28	1300	0.006307	-2.151	1.727	-0.4237
Run 29	1300	0.006473	-2.208	1.773	-0.4349
Run 30	1300	0.006639	-2.264	1.818	-0.446
Run 31	1300	0.006805	-2.321	1.864	-0.4572
Run 32	1300	0.006971	-2.377	1.909	-0.4683
Run 33	1300	0.007137	-2.434	1.955	-0.4795
Run 34	1300	0.007303	-2.491	2	-0.4906
Run 35	1300	0.007469	-2.547	2.045	-0.5018
Run 36	1300	0.007635	-2.604	2.091	-0.5129
Run 37	1300	0.007801	-2.66	2.136	-0.5241
Run 38	1300	0.007967	-2.717	2.182	-0.5352
Run 39	1300	0.008133	-2.774	2.227	-0.5464
Run 40	1300	0.008299	-2.83	2.273	-0.5575
Run 41	1300	0.008465	-2.887	2.318	-0.5687
Run 42	1300	0.008631	-2.943	2.364	-0.5798
Run 43	1300	0.008797	-3	2.409	-0.591
Run 44	1300	0.008963	-3.057	2.455	-0.6021
Run 45	1300	0.009129	-3.113	2.5	-0.6133
Run 46	1300	0.009295	-3.17	2.545	-0.6244
Run 47	1300	0.009461	-3.227	2.591	-0.6356
Run 48	1300	0.009627	-3.283	2.636	-0.6468
Run 49	1300	0.009793	-3.34	2.682	-0.6579
Run 50	1300	0.009959	-3.396	2.727	-0.6691

Parametric Table: Table 2

	i	$\dot{m}$	Q_{Cond} [kW]	Q_{Evap} [kW]	W_{Comp} [kW]
Run 51	1300	0.01012	-3.453	2.773	-0.6802
Run 52	1300	0.01029	-3.51	2.818	-0.6914
Run 53	1300	0.01046	-3.566	2.864	-0.7025
Run 54	1300	0.01062	-3.623	2.909	-0.7137
Run 55	1300	0.01079	-3.679	2.955	-0.7248
Run 56	1300	0.01095	-3.736	3	-0.736
Run 57	1300	0.01112	-3.793	3.045	-0.7471
Run 58	1300	0.01129	-3.849	3.091	-0.7583
Run 59	1300	0.01145	-3.906	3.136	-0.7694
Run 60	1300	0.01162	-3.962	3.182	-0.7806
Run 61	1300	0.01178	-4.019	3.227	-0.7917
Run 62	1300	0.01195	-4.076	3.273	-0.8029
Run 63	1300	0.01212	-4.132	3.318	-0.814
Run 64	1300	0.01228	-4.189	3.364	-0.8252
Run 65	1300	0.01245	-4.245	3.409	-0.8363
Run 66	1300	0.01261	-4.302	3.455	-0.8475
Run 67	1300	0.01278	-4.359	3.5	-0.8586
Run 68	1300	0.01295	-4.415	3.545	-0.8698
Run 69	1300	0.01311	-4.472	3.591	-0.8809
Run 70	1300	0.01328	-4.528	3.636	-0.8921
Run 71	1300	0.01344	-4.585	3.682	-0.9032
Run 72	1300	0.01361	-4.642	3.727	-0.9144
Run 73	1300	0.01378	-4.698	3.773	-0.9255
Run 74	1300	0.01394	-4.755	3.818	-0.9367
Run 75	1300	0.01411	-4.811	3.864	-0.9478
Run 76	1300	0.01427	-4.868	3.909	-0.959
Run 77	1300	0.01444	-4.925	3.955	-0.9701
Run 78	1300	0.01461	-4.981	4	-0.9813
Run 79	1300	0.01477	-5.038	4.045	-0.9924
Run 80	1300	0.01494	-5.094	4.091	-1.004
Run 81	1300	0.0151	-5.151	4.136	-1.015
Run 82	1300	0.01527	-5.208	4.182	-1.026
Run 83	1300	0.01544	-5.264	4.227	-1.037
Run 84	1300	0.0156	-5.321	4.273	-1.048
Run 85	1300	0.01577	-5.378	4.318	-1.059
Run 86	1300	0.01593	-5.434	4.364	-1.07
Run 87	1300	0.0161	-5.491	4.409	-1.082
Run 88	1300	0.01627	-5.547	4.455	-1.093
Run 89	1300	0.01643	-5.604	4.5	-1.104
Run 90	1300	0.0166	-5.661	4.545	-1.115
Run 91	1300	0.01676	-5.717	4.591	-1.126
Run 92	1300	0.01693	-5.774	4.636	-1.137
Run 93	1300	0.0171	-5.83	4.682	-1.149
Run 94	1300	0.01726	-5.887	4.727	-1.16
Run 95	1300	0.01743	-5.944	4.773	-1.171
Run 96	1300	0.01759	-6	4.818	-1.182
Run 97	1300	0.01776	-6.057	4.864	-1.193
Run 98	1300	0.01793	-6.113	4.909	-1.204
Run 99	1300	0.01809	-6.17	4.955	-1.215
Run 100	1300	0.01826	-6.227	5	-1.227


R32
P [kPa]
h [kJ/kg]
10^4
10^3
10^2
10^1
10^0
10^{-1}
5×10^{-2}
-100
0
100
200
300
400
500
600
0.2
0.4
0.6
0.8


R32
6x10^3
10^3
10^2
P [kPa]
200
300
400
500
h [kJ/kg]


Temp
i
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15
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21
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27
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39
42
45
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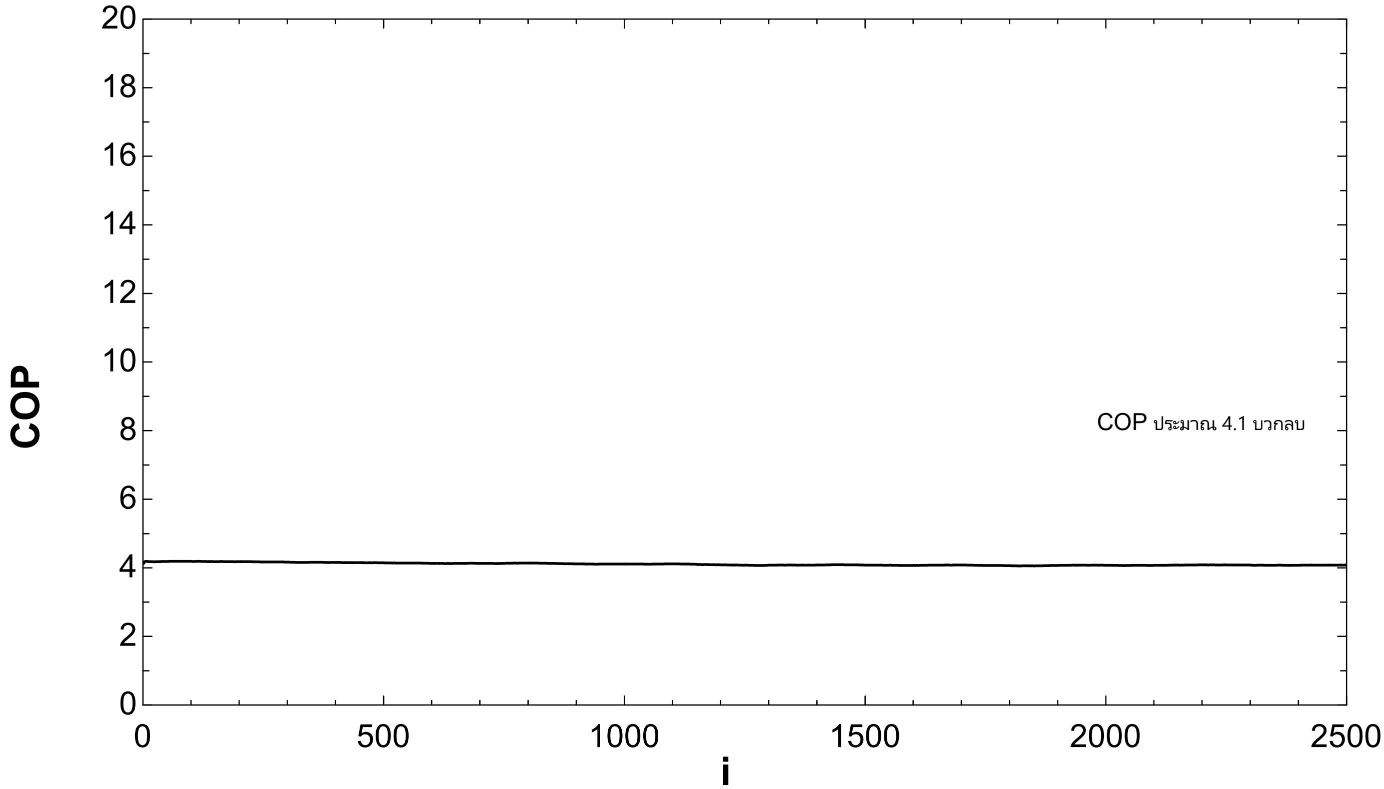
COP
i
COP ประมาณ 4.1 บวกลบ
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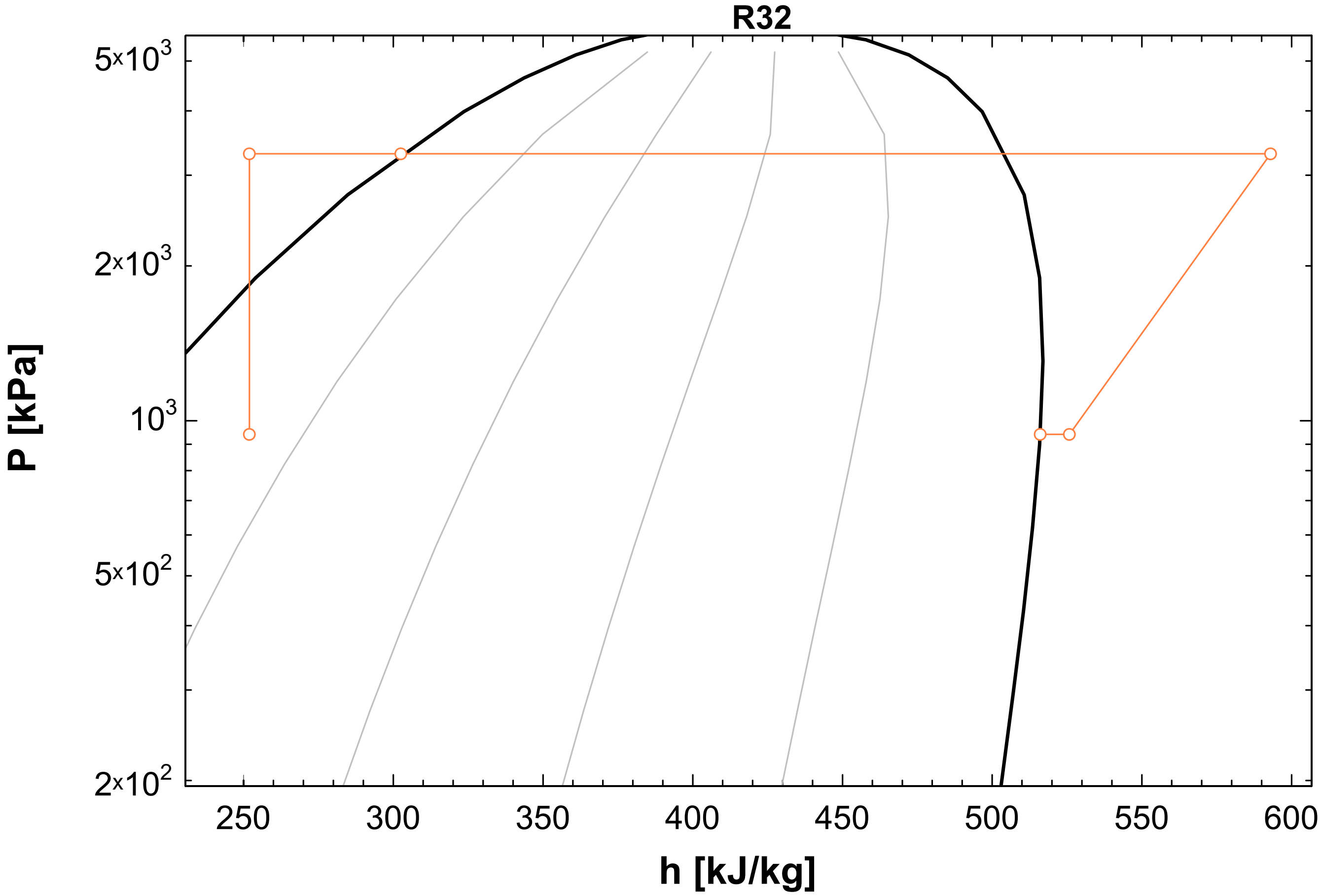
1000
1500
2000
2500
0
2
4
6
8
10
12
14
16
18
20


R32
5x10^3
2x10^3
10^3
5x10^2
2x10^2
P [kPa]
250
300
350
400
450
500
550
600
h [kJ/kg]